

House Price Prediction and Analysis

Project Summary

The objective of this project was to analyze housing data and build machine learning models capable of predicting house prices based on various property characteristics. The dataset consisted of 545 house records containing features such as area, number of bedrooms, bathrooms, stories, parking spaces, furnishing status, air conditioning, and location-related factors.

The project began with data exploration and cleaning. The dataset was checked for missing values, duplicate records, and data quality issues. No missing values or duplicate records were found, making the dataset suitable for analysis. Categorical variables such as furnishing status, air conditioning, guest room availability, and preferred area were converted into numerical format using one-hot encoding so that machine learning algorithms could process them effectively.

Exploratory Data Analysis (EDA) was performed to understand the relationships between different features and house prices. Visualizations such as histograms and correlation heatmaps revealed that property area, number of bathrooms, air conditioning, number of stories, and parking availability were among the most influential factors affecting house prices. Larger homes with better amenities generally commanded higher market values.

Two machine learning models were developed and compared: Linear Regression and Random Forest Regressor. The dataset was divided into training and testing sets using an 80:20 ratio. Both models were trained on the training data and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

The results showed that the Linear Regression model slightly outperformed the Random Forest model. Linear Regression achieved an R^2 score of approximately 0.65, meaning it explained around 65% of the variation in house prices. This indicates that the selected features provide a reasonably strong basis for predicting property values. The Random Forest model also performed well but did not surpass the Linear Regression model on this dataset.

Overall, this project demonstrated how data analytics and machine learning can be used to estimate house prices and identify the factors that drive property values. The findings suggest that real estate businesses should focus on larger properties with modern amenities and desirable locations, as these characteristics contribute most significantly to higher house prices.